



AccessMap

Can data predict which sidewalks are accessible?

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Live demo: accessmap-13a.streamlit.app

The Problem

Wheelchair users can't tell if a sidewalk is passable before they get there.

Too narrow, curb too high — they find out on the spot. A wrong route isn't an inconvenience; it can end a trip.

Maps show that a road exists. They don't show whether a wheelchair can use it.

Research question: Can data predict which sidewalks are accessible for wheelchair users?

Why it matters



Wheelchair users, caregivers, seniors plan routes with confidence



Cities collect sidewalk data but rarely turn it into anything usable



A working model plugs into any routing app as a scoring layer

Two Datasets, One Question



PMR — Amsterdam sidewalks

72,274

measured segments

Physical measurements: width, curb height, crossings. Measures the physical world.



Housing — US addresses

10,104

crowd-labeled rows

Crowd-sourced “sidewalk ok?” labels judged from Google Maps street images.

No geographic overlap. One is meters in Amsterdam, the other spans a continent in degrees. Same pipeline runs on each; results are compared at the model level. Rows are never merged.

The Algorithms

Random Forest (main model)

Type: Supervised classification — many decision trees vote

Inputs: PMR: width, curb height, crossing, length. Housing: house type + crowd-confidence scores

Output: accessible / not accessible, with a probability

Why we chose it (pros)

- Handles mixed features well
- Feature importance is readable — that's how we caught the bias
- Strong baseline without heavy tuning

Trade-offs (cons)

- Can't invent signal that isn't in the features
- Happily rides a biased shortcut feature
- Less interpretable than a single tree

Supporting tools

ModernBERT

Embeds each address into a 768-dim vector (6,425 addresses)

FAISS

Similarity search over those vectors: “find addresses like this one,” ranked by distance

How the Project Evolved

V1

Baselines

Both models trained. Two problems surface: Housing below baseline, PMR over-flags.

V2

Bias fix

Drop state from Housing, retrain. Result: a null finding — and an honest one.

V3

Precision fix

PMR threshold tuning fails. Root cause found in features; curb height added.

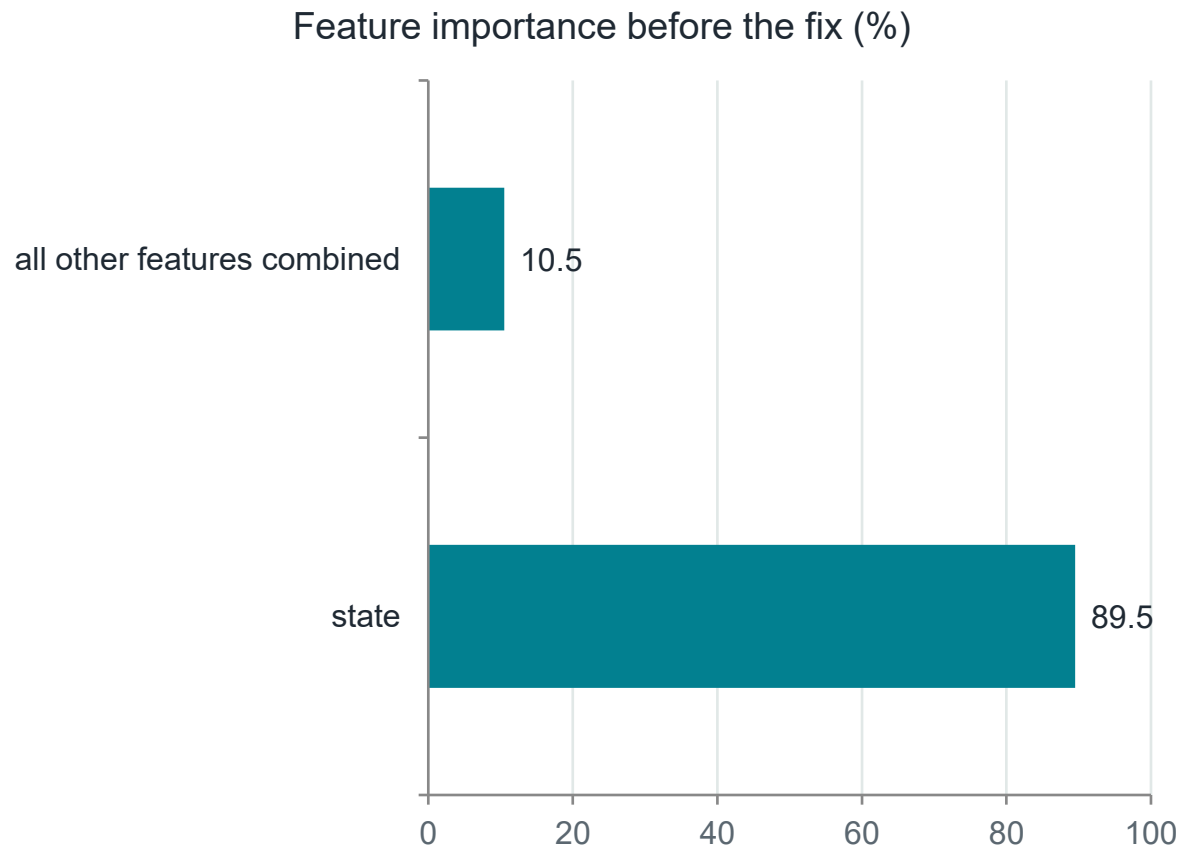
V4

Deployment

Streamlit app built, tested locally + on the cloud, live for anyone to use.

We didn't get it right the first time — and that's the story worth telling.

Challenge 1 — The Housing Model Learned Geography

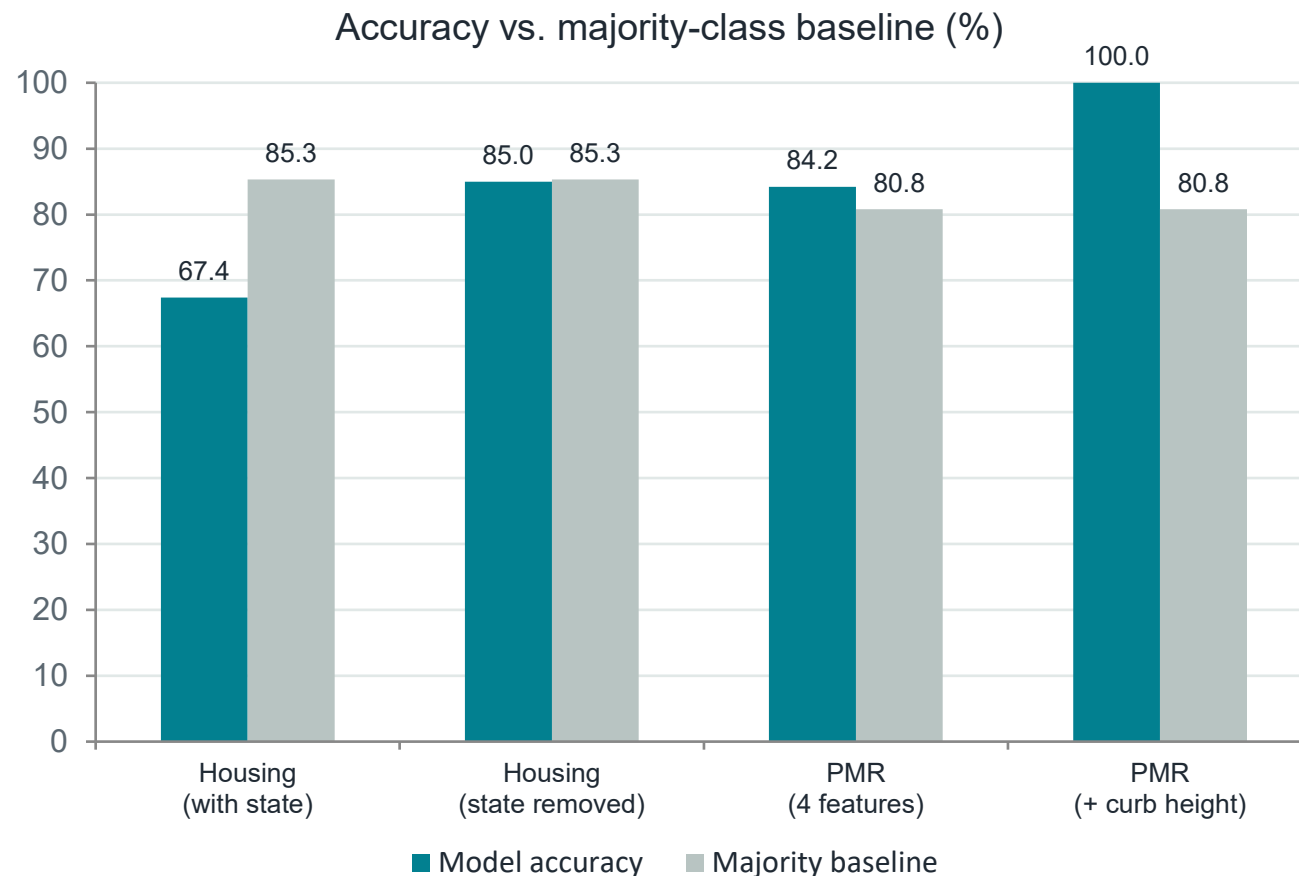


67.4% accuracy vs. an 85.3% baseline — worse than guessing.

One feature, **state**, drove 89% of every decision. The model wasn't judging house condition. It was memorizing which states have more “bad sidewalk” labels.

This is a bias problem, not a tuning problem.

The Fix Revealed the Real Finding



Housing: 85.0% ties the 85.3% baseline. Recall on bad sidewalks: 0.01. Without state, no signal is left — and the raw data has no other features to add.

A null result is a result. “This dataset can't predict sidewalk condition, and here's why” is our headline bias finding — not a failure to hide.

Challenge 2 — Confidently Wrong, Not Unsure



Attempt 1: move the threshold — failed

PMR over-flagged “not accessible” (precision 0.55). We raised the cutoff from 0.5 to 0.65. Nothing changed.

14 / 14,455

test predictions sat anywhere near the cutoff. Probabilities were pinned at 0 or 1 — the model wasn't hesitating, it was confidently wrong. A threshold can't fix confident mistakes.



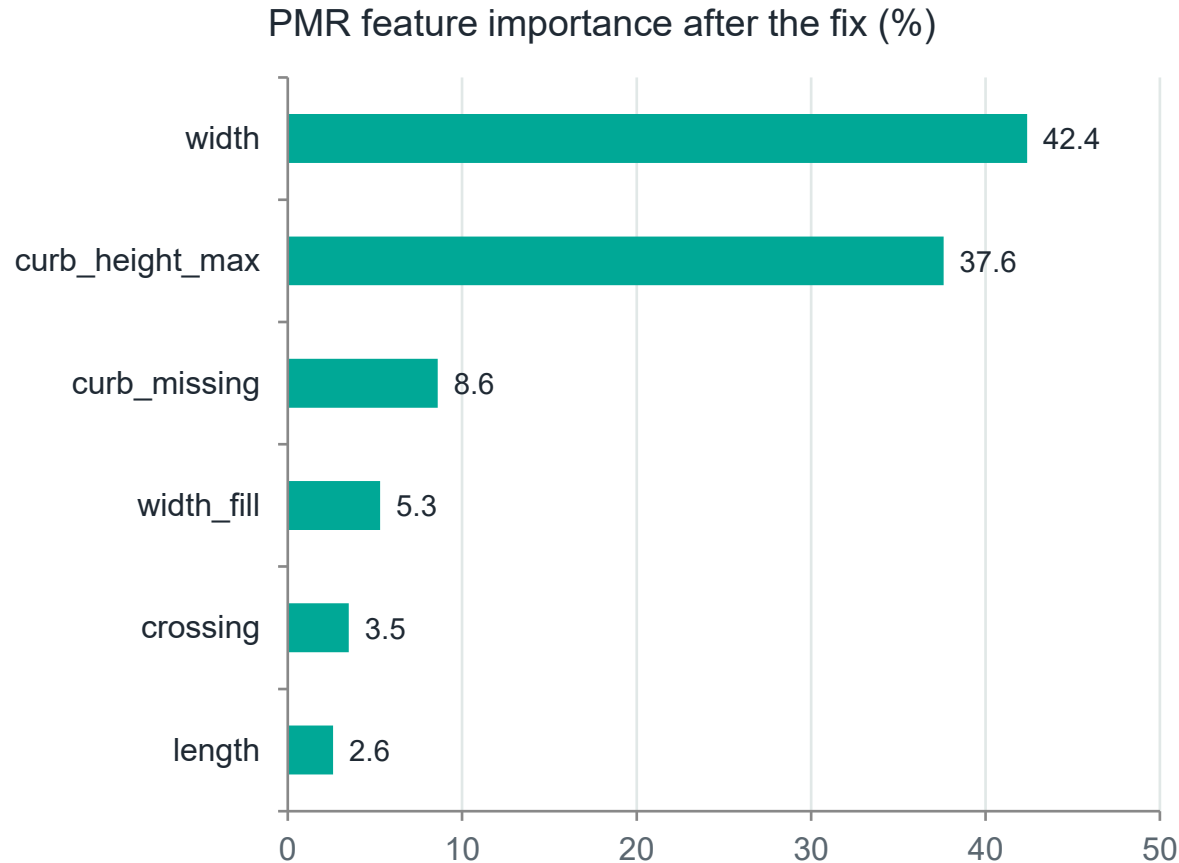
Attempt 2: fix the features — worked

Our label rule = width + curb height. But the model only ever saw width — it was guessing the curb half blind.

We added curb_height_max and a missing-value flag (74% of rows have no curb measurement — the missingness itself is information).

Lesson: check the probability distribution before reaching for a threshold.

Evaluation — Read the Numbers Honestly



PMR: 100% accuracy.

Expected, not impressive. The two label inputs (width + curb height) now top the chart — the model recovers the rule we wrote. It proves the pipeline is consistent, nothing more.

Housing: 85.0%.

That's the majority-class rate, not skill. The honest conclusion stands: house-level features alone can't predict sidewalk condition.

Live Demo

accessmap-13a.streamlit.app



Sidewalk Checker

Enter width, curb height, crossing, length →
accessible / not accessible + confidence



Address Search

Pick an address → FAISS returns similar
addresses with sidewalk labels + distances



About & Limits

What the app does, what it refuses to do,
and why — honesty as a feature

One deployment decision we're proud of: the app does NOT predict sidewalk condition for US addresses. We tested that model — it has no signal — so the app reports the real crowd label instead of pretending to predict.

Impact — Both Sides



Positive

- Wheelchair users and caregivers plan routes before leaving home
- Cities can find clusters of bad segments and prioritize repairs
- Accessibility becomes a data layer any routing app can adopt



Negative / risks

- A biased model could systematically mislabel whole regions — exactly what our Housing model did before the fix
- False “accessible” labels are worse than no label: someone plans a trip they can't finish
- Rule-based labels (ours) can encode wrong assumptions if never validated against real users

Essential Question

How can the process of creating AI/ML solutions amplify or mitigate bias in our case?

It amplified bias when...

we let a geographic proxy (state) dominate training. The model looked like it was scoring house condition; it was really scoring ZIP codes. High feature importance made an unfair shortcut look like intelligence.

It mitigated bias when...

we audited feature importance, removed the proxy, retested, and reported the honest null result instead of a prettier number. The same ML process that created the bias also gave us the tools to catch it — if we chose to look.

Next Steps



Rule-free PMR experiment

Retrain on only features outside the label rule (crossing, length, width_fill): can accessibility be guessed without measuring? The one test where the model can show real skill.

Week 13



Validate the label

Compare our width/curb rule against the 50 verified accessibility labels; report agreement.

Week 13



Location-aware retrieval

Add coordinates to the PMR index so retrieval ranks within tied groups instead of returning exact ties.

Week 14



Individual roadmaps

Each member continues per their My AI Roadmap; repo stays public as a portfolio piece.

Ongoing

Citations & Sources

Data sources

- City of Amsterdam Open Data — PMR sidewalk accessibility dataset (72,274 segments)
- CrowdFlower / Figure Eight “Data for Everyone” — Housing: Wheelchairs dataset (10,104 crowd-labeled US addresses)

Related work & methods

- Breiman, L. (2001). Random Forests. Machine Learning, 45(1).
- Johnson, J., Douze, M., & Jégou, H. (2019). Billion-scale similarity search with GPUs. IEEE Transactions on Big Data. (FAISS)
- Warner, B., et al. (2024). Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder. (ModernBERT)
- Mehrabi, N., et al. (2021). A Survey on Bias and Fairness in Machine Learning. ACM Computing Surveys.
- Saha, M., et al. (2019). Project Sidewalk: A Web-based Crowdsourcing Tool for Collecting Sidewalk Accessibility Data at Scale. ACM CHI.

Live app: accessmap-13a.streamlit.app Code + docs: github.com/zhang-systems/ai4all-accessmap



Thank You!

Questions? Try the app yourself while we talk.

accessmap-13a.streamlit.app

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